

Welcome to Code Crunch!

C12 | Crunch 3D: CAD & Modeling

[Tues 4-5PM] Weekly | Spring 2025

Attendance

Earn a Certificate of
Completion by attending
at least 70% of the
workshops



CODECRUNCH.CS.FIU



305 Hackathon : Edition Spring 2025



305 

CODE CRUNCH

HACKATHON **HYBRID**

MAR 24 - 27 & MAR 29 - 31, 2025
WORKSHOPS | COMPETE | IN-PERSON CAREER FAIR

FIU Honors College **epichire**

SIGN UP HERE: [CODECRUNCH.CS.FIU.EDU](https://codecrunch.cs.fiu.edu)

FIU ORGS COLLABORATORS

**OPEN TO ALL STUDENTS, REGARDLESS OF MAJOR
—NO EXPERIENCE NEEDED! FRESHMEN,
GRADUATES, AND ALUMNI FROM ANY UNIVERSITY
ARE WELCOME. LET'S INNOVATE TOGETHER!**

#CODECRUNCH-HACKS

Funded by SGA
| Open to All University Students

Engineering Research Society
FLORIDA ENGINEERING SOCIETY
AI & FIU
THE NATIONAL SOCIETY OF BLACK ENGINEERS
PANTHER ROBOTICS
IBHS
AM>|FIU
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FIU SHPE
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Spring 2025 Weekly Schedule





Funded by SGA | Open to All Students

CODE CRUNCH WEEKLY SCHEDULE - SPRING 25

Your CS Club for Collaboration, Networking, and Hands-On Projects!
Open to all students, regardless of major—no experience needed! **Freshmen, graduates, and alumni from any university are welcome.** Let's innovate together!

C5 Crunch AI Data Science Python MON 1 - 2 PM WED 12:30 - 1:30 PM	C3 CodeCrunch LABS Portfolio Java & React MON 2 - 3 PM	C7 Wire Crunch: Embedded Systems C++ TUE 1 - 3 PM	C11 Crunch Arcade Unity TUE 3 - 4 PM FRI 4 - 5 PM	C12 Crunch 3D: CAD & Modeling OnShape TUE 4 - 5 PM	C9 Swe Convos Networking WED 1:30 - 2:30 PM FRI 2 - 3PM ***
C1 Crunch Convos Python WED 2:30 - 3:30 PM	C2 Crunch The Code Python WED 3:30 - 4:30 PM	C6 Cybersecurity Crunch VM/LINUX THU 1 - 2 PM	C4 CodeCrunch LABS Hackathon Java & React THU 2 - 3 PM	C8 CodeCrunch LABS: Web Dev Javascript THU 3 - 4 PM	C10 Crunching Bioinformatics Python FRI 3 - 4 PM

Attend 7 of 10 Weekly Sessions in the Same Workshop to Earn a Certificate of Completion!

RSVP on
LU.MA/CODECRUNCH

In person or
Stream on
Zoom



 CODECRUNCH.CS.FIU.EDU

 **January 6, 2025 - April 4, 2025**
Cost: No-cost

Become a member



Unit #5

Topic: Final Project

C12 | Crunch 3D: CAD & Modeling

Key Dates and Structure

C12 - Crunch 3D: CAD & Modeling



- ★ **Week 1: Introduction to OnShape, Sketching Fundamentals, & 2D Constraints**

Learning what On-Shape is and how to properly sketch on OnShape.

- ★ **Week 2: 3D Part Modeling**

Learning how to use extrude, remove, fillet, chamfer, revolve, shell, and loft functions for 3D models.

- ★ **Week 3: 3D Part Modeling Continued**

Learning how to use draft, mirror, and sweep functions for 3D models.

- ★ **Week 4: Assemblies & Mating Constraints**

Learning how to create assemblies through mates and constraints.

- ★ **Week 5: Final Project - Real-World Design Challenge**

Applying all concepts to design a functional product.

- ★ **Week 6: Final Project Showcase**

Presenting your final project, sharing your experience, and receiving feedback.

Session Overview

In this session we will be working on our **final project!**
We will be putting all the skills we have learned to use.

You can create **anything** that you desire. Here are a few ideas:

- Drone frame
- Robotic arm component
- Phone holder
- 4-cylinder piston
- Mini windmill
- Handcar
- Toy car

You can follow along with use as we create a **mini windmill!**



Time for...

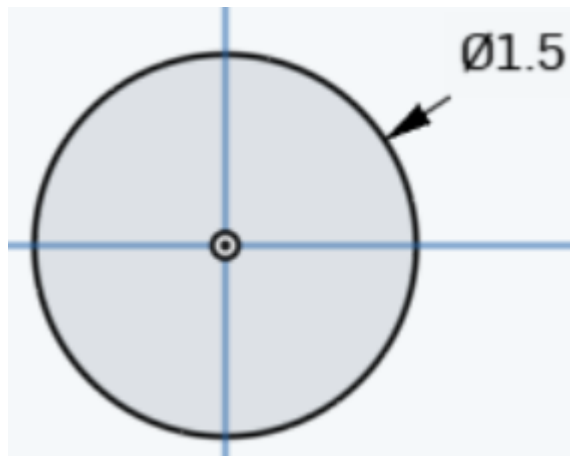


Part 1 – The Pole!

Part 1: Loft Set Up

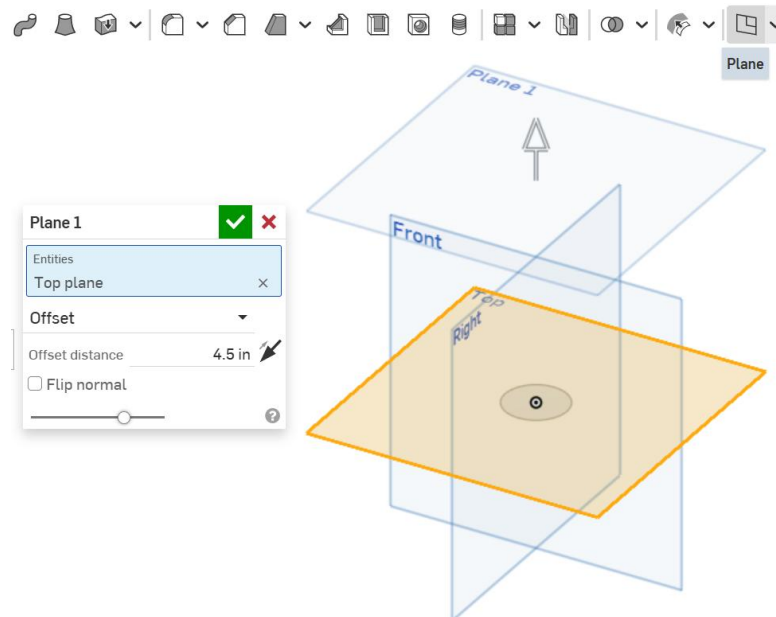
Step 1:

Make a **sketch** on the **Top Plane** with a **Center Point Circle** of diameter 1.5 in.



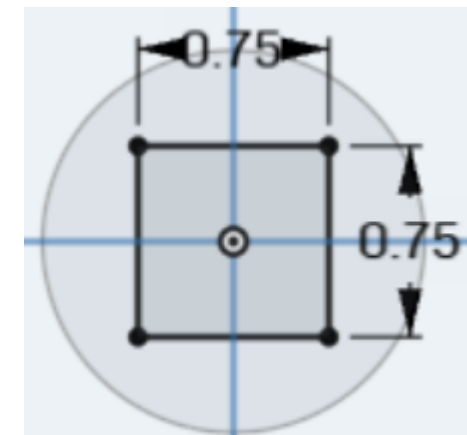
Step 2:

Make an offset **Plane** of the **Top Plane** with a distance of 4.5 in.



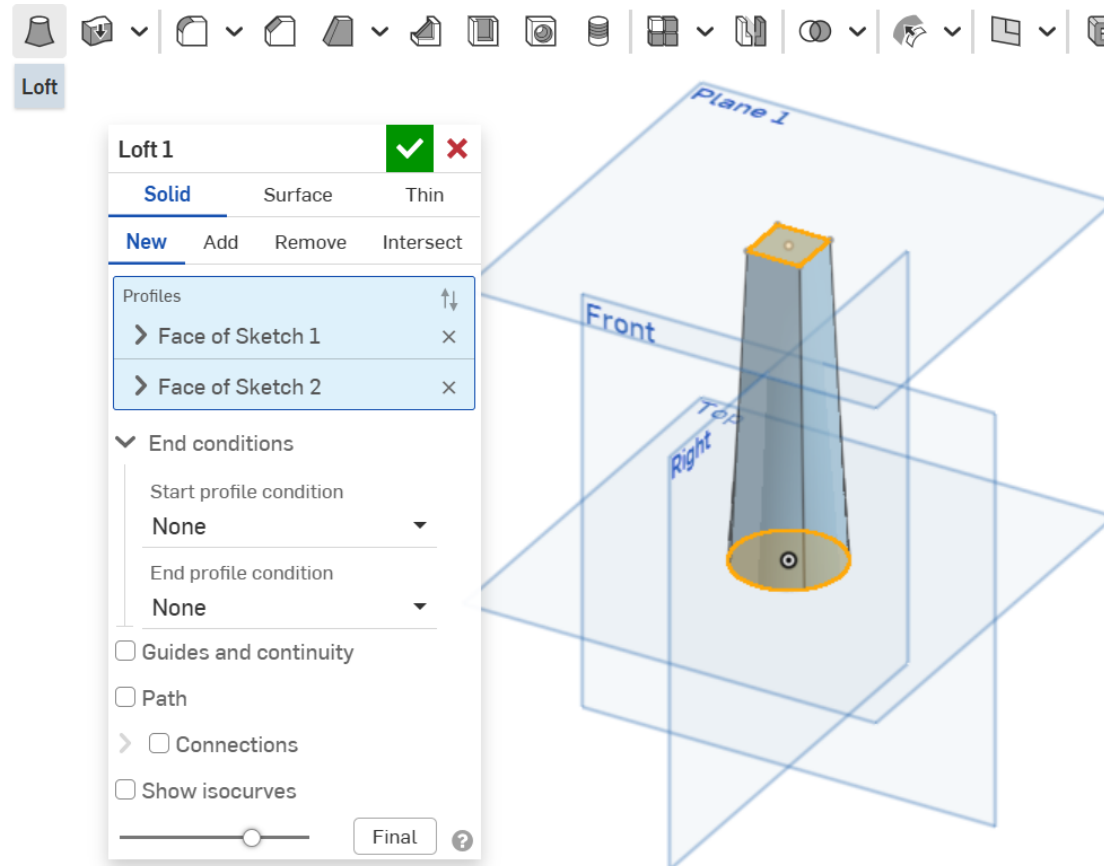
Step 3:

Make a **sketch** on the **Plane 1** with a **Center Point Rectangle** that is 0.75 in by 0.75 in.



Part 1: Loft

Make a **Loft** using **Sketch 1** and **Sketch 2**.

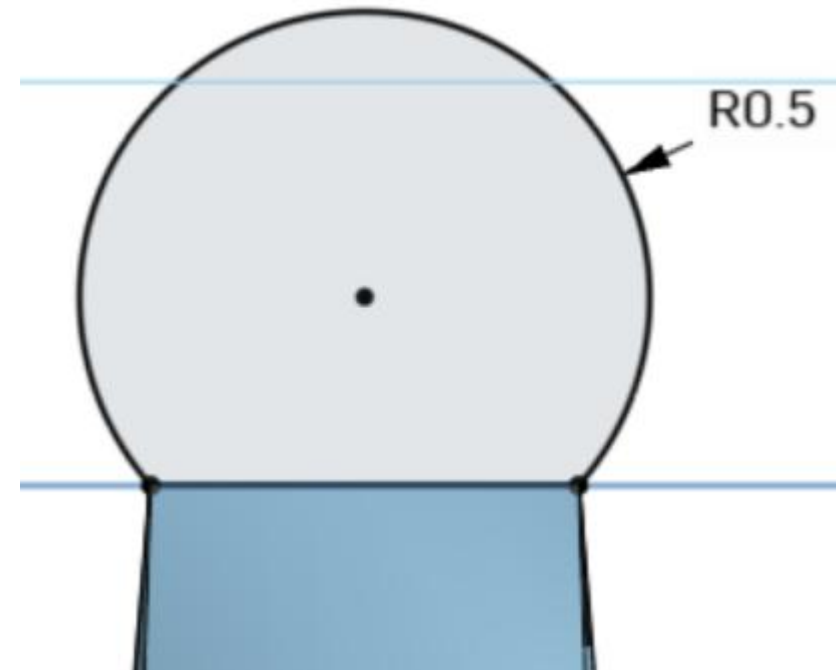


Part 1: Sketch 3

Make **Sketch 3** on the **Front Plane**.

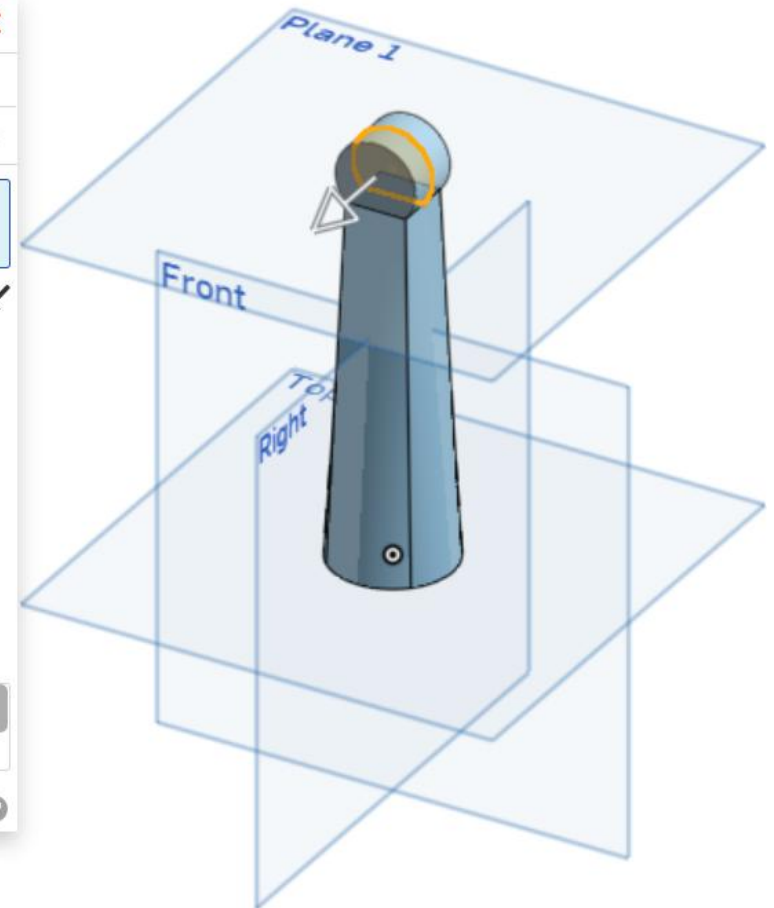
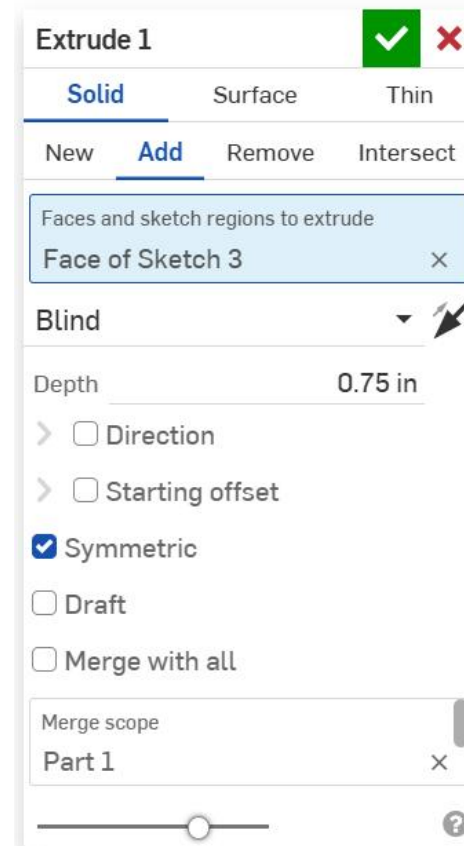
Click to **Use** the top line of the pole base in this sketch.

Then draw a **3 Point Arc** – the first 2 points touching the ends of the line and the 3rd point being placed somewhere above the line. Then change the **radius** of the arc to be 0.5 in.



Part 1: Extrude

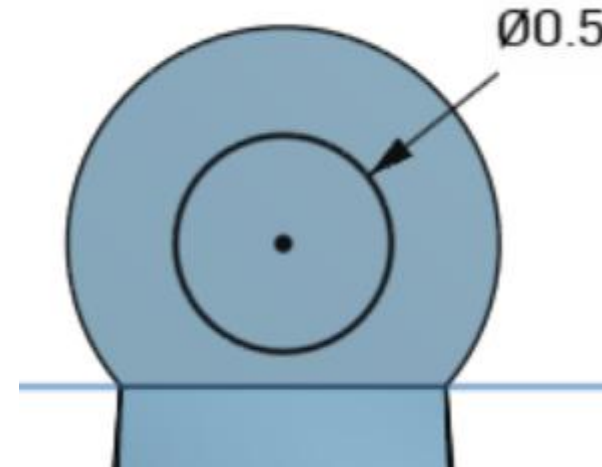
Make a **Symmetric Extrude** of **Sketch 3** with a depth of 0.75 in.



Part 1: Sketch 4

Make **Sketch 4** on the **Front Face** of the part that was just extruded.

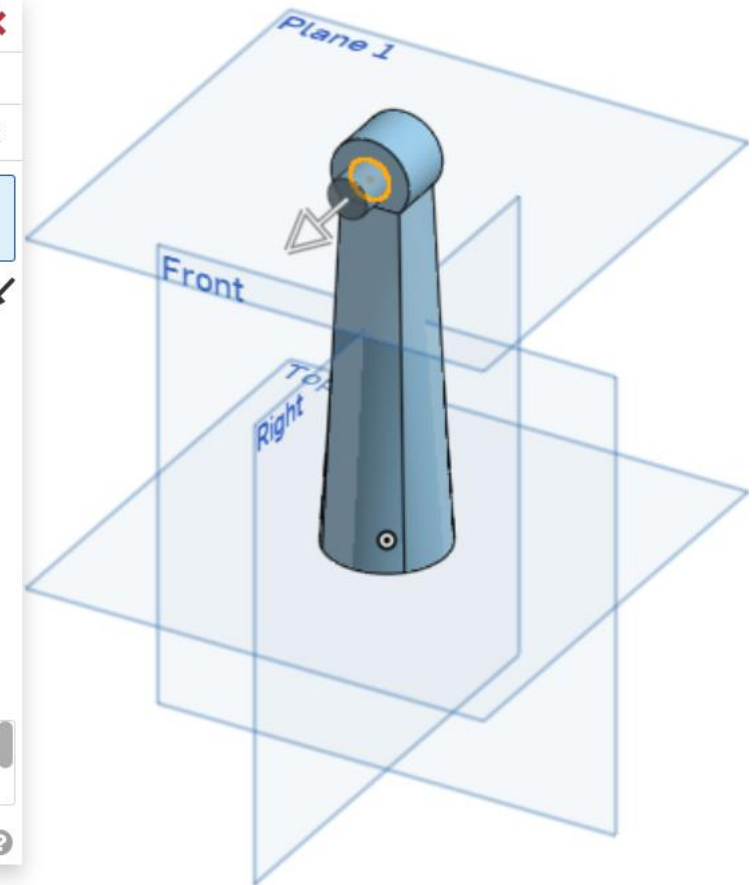
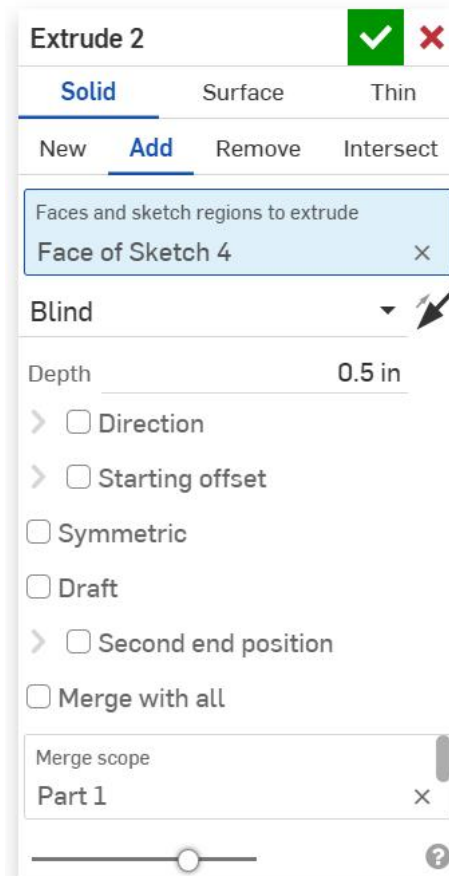
Draw a **Center Point Circle** on the center of the arc of the model, with a **diameter** of 0.5 in.



Part 1: Extrude

Make an **Extrude** of **Sketch 4** with a **depth** of 0.5 in.

The pole is now finished!



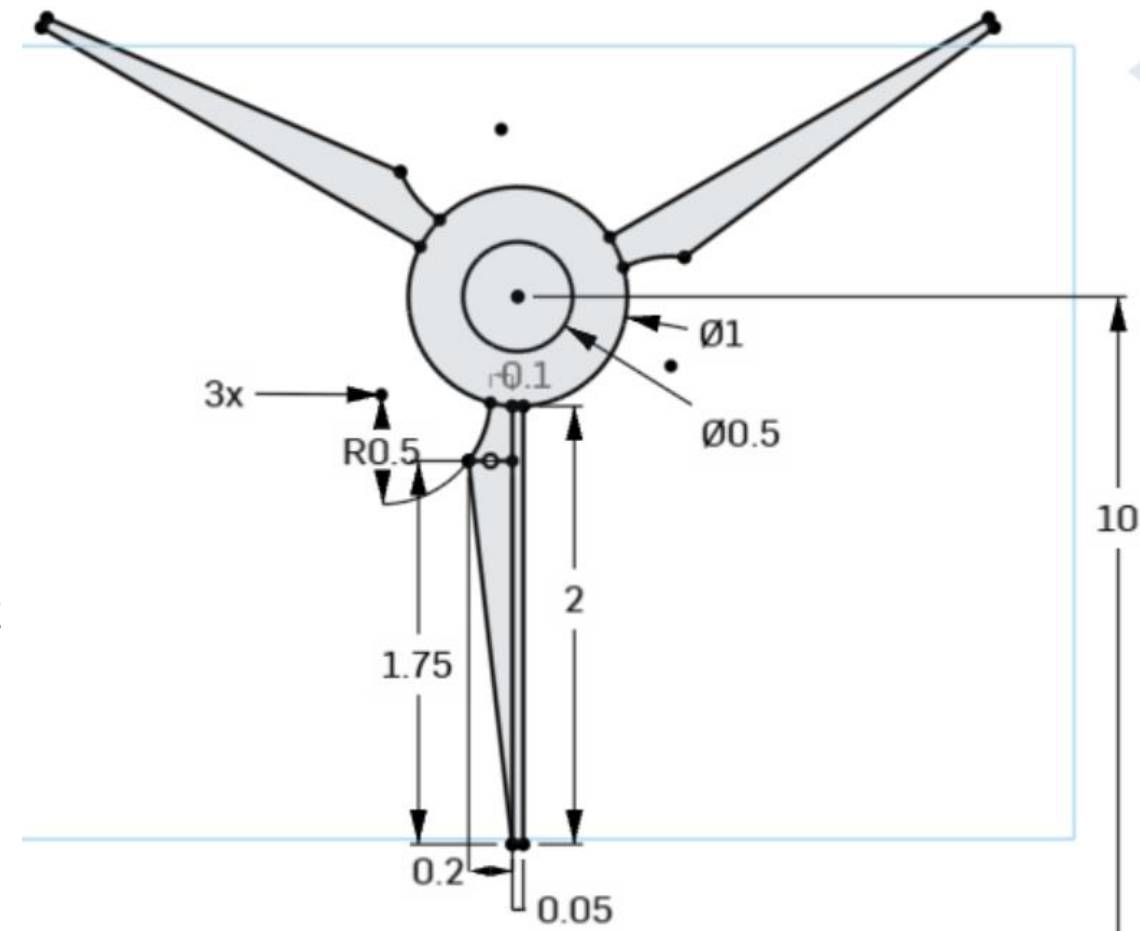
Time for...



Part 2 – The Blades!

Part 2: Sketch 5

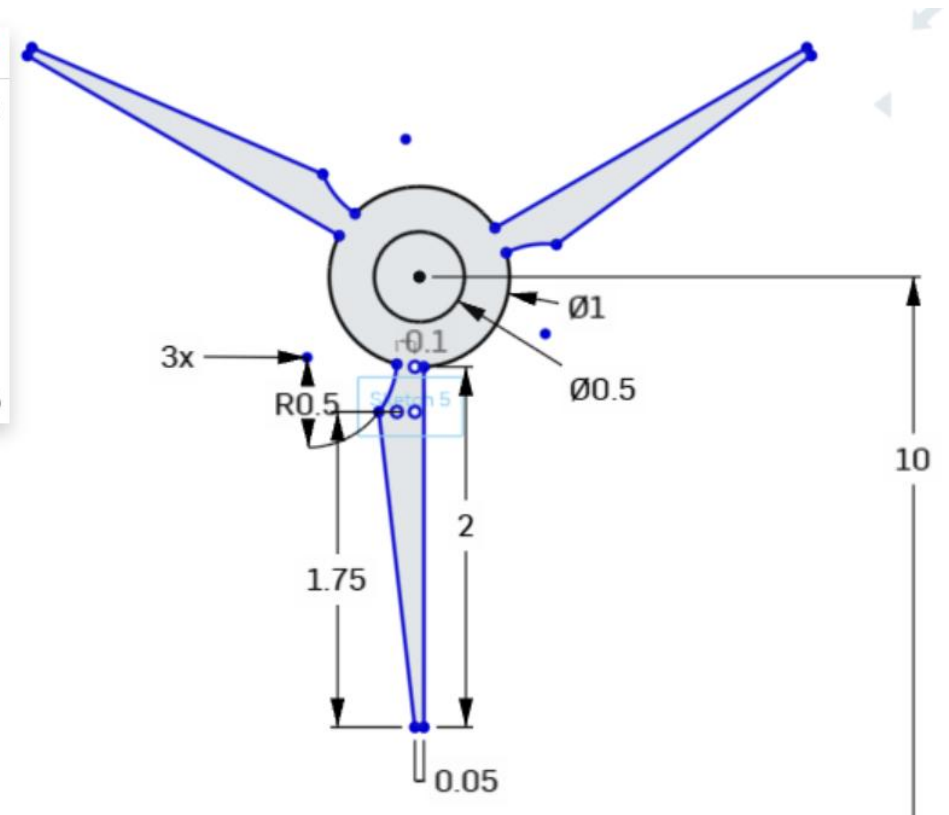
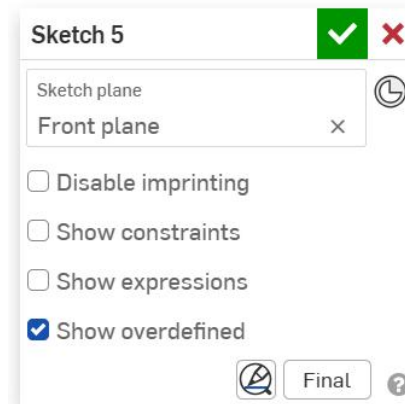
1. Make **Sketch 5** on the **Front Plane**.
2. Start with a **Center Point Circle** with a **diameter** of 0.5 in. Place the center of this circle 10 in away from the origin. Make another **Center Point Circle** of **diameter** of 1 in that shares the same center.
3. Next create a blade using a **Corner Rectangle** of 0.05 in by 2 in, whose ends **coincide** with the larger circle. Then create a **Triangle** at the end of the blade with a base of 0.2 in and height of 1.75 in. Then create a **3 Point Arc** with a **radius** of 0.5 in to connect the triangle to the larger circle. Make the **distance** between the points on the larger circle of the arc and of the left corner of the rectangle to be 0.1 in apart.
4. Use a **Circular Pattern** to replicate the blade 3x around the center of the circles.



Part 2: Trim

Trim all the lines in the sketch that are unnecessary. This will ensure that when we extrude, the extrusion will produce one part.

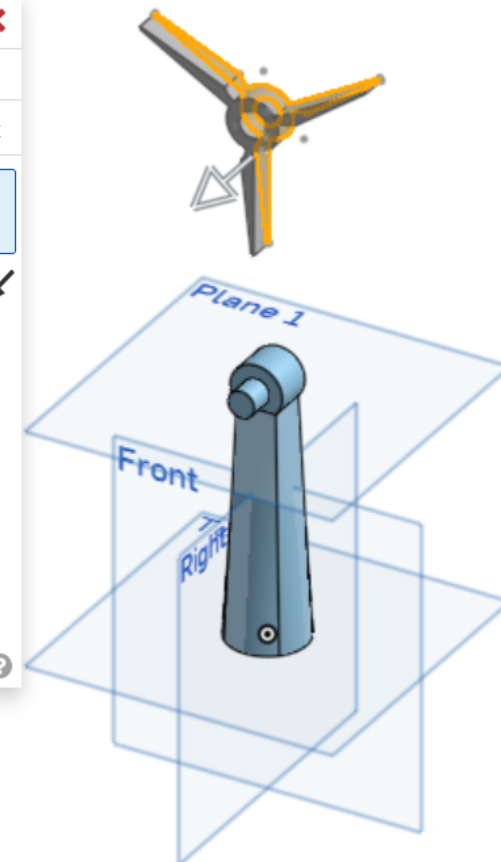
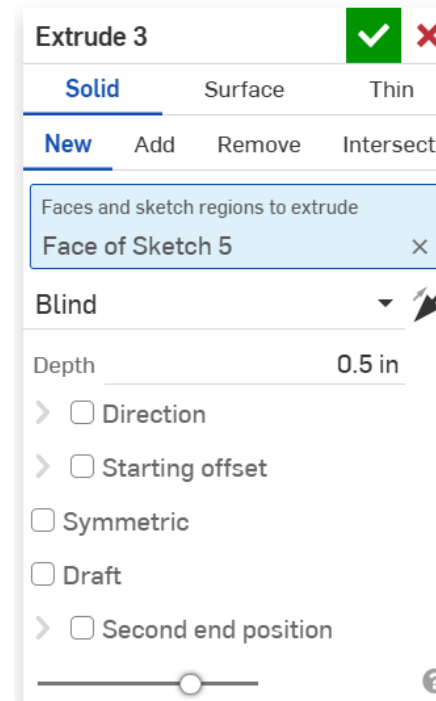
Note: This will make the sketch not appear as fully defined anymore. If you wish, you can add the dimensions again.



Part 2: Extrude

Make an **Extrude** of **Sketch 5** with a **depth** of 0.5 in.

The blades are now finished!

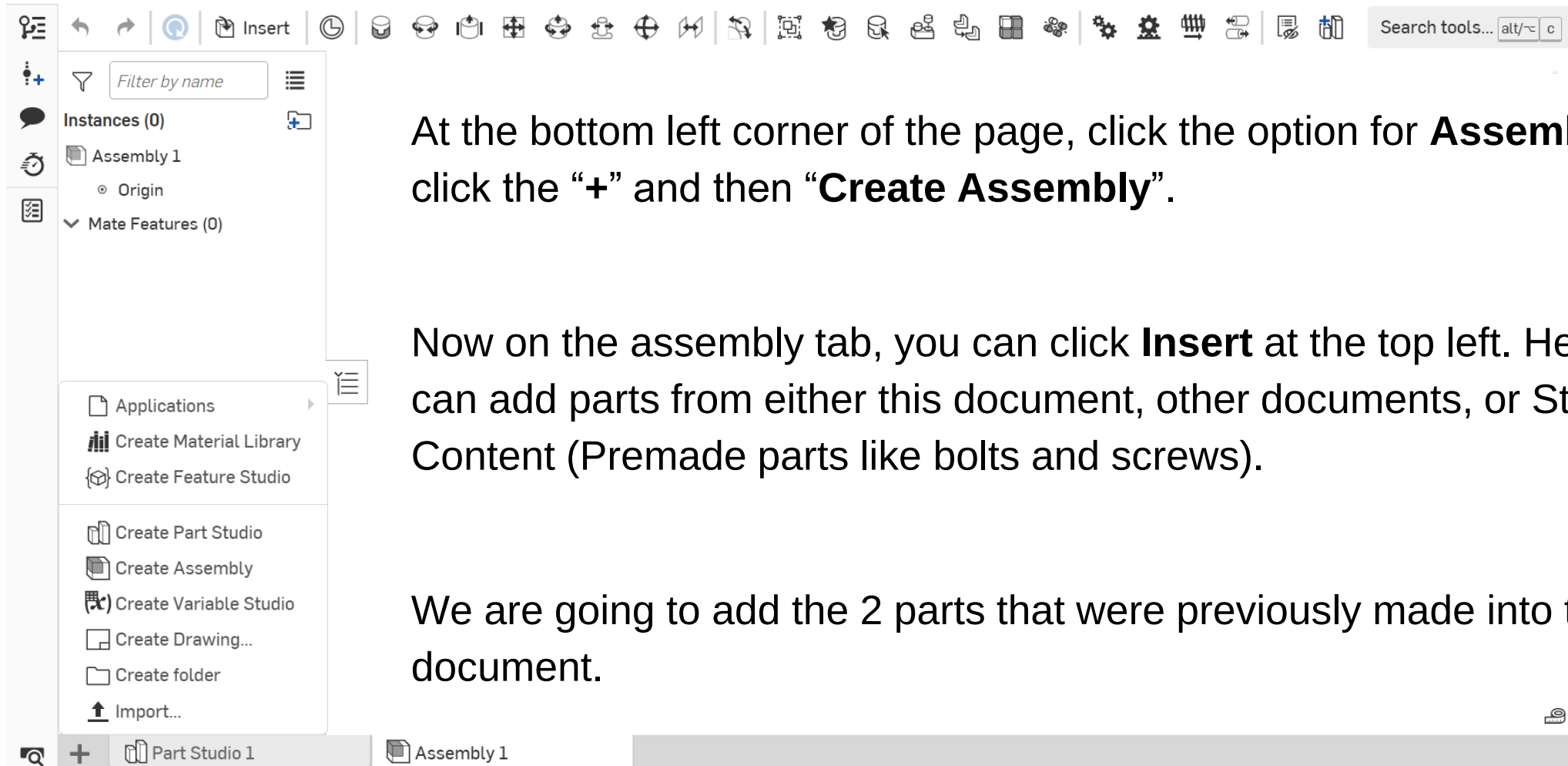




Time for the...

Windmill Assembly!

Assembly



At the bottom left corner of the page, click the option for **Assembly 1** or click the “+” and then “**Create Assembly**”.

Now on the assembly tab, you can click **Insert** at the top left. Here you can add parts from either this document, other documents, or Standard Content (Premade parts like bolts and screws).

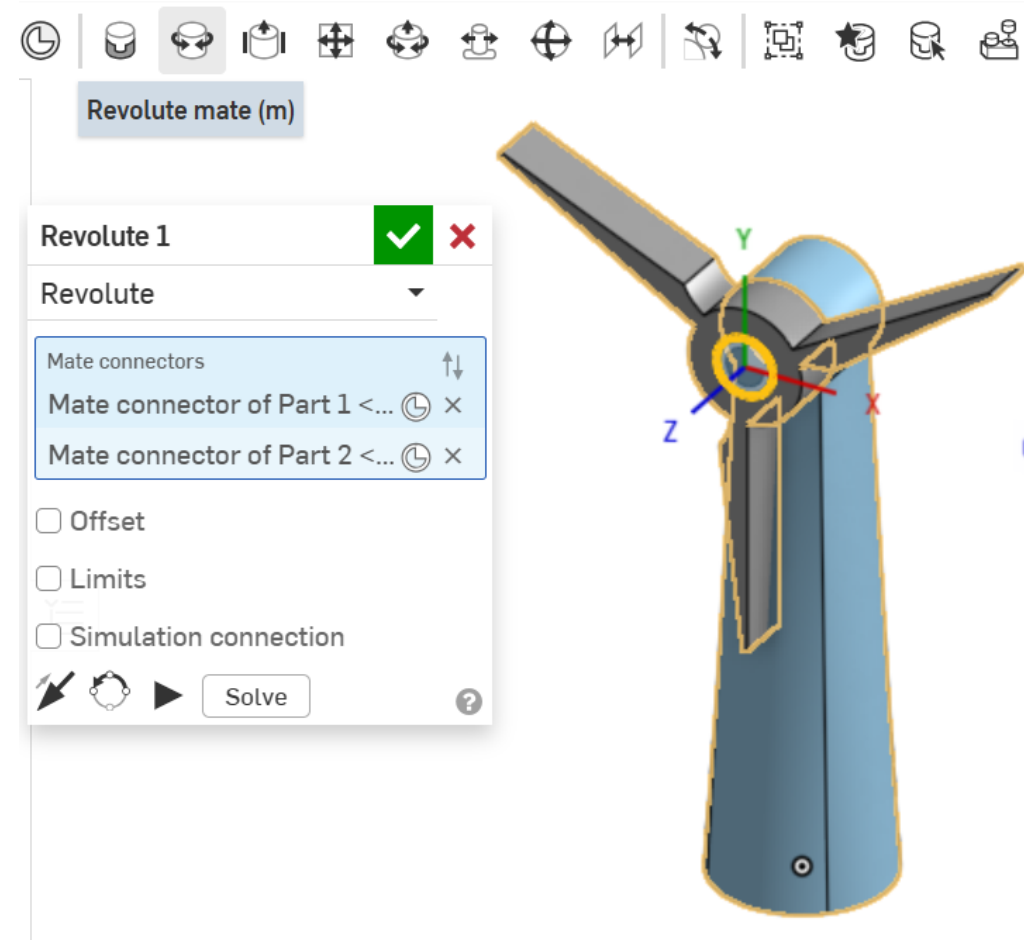
We are going to add the 2 parts that were previously made into this document.

Windmill: Assembly

To create the assembly, start off by making **Part 1** the “**base**”. To do so, right click **Part 1** and select **Fix**.  Fix

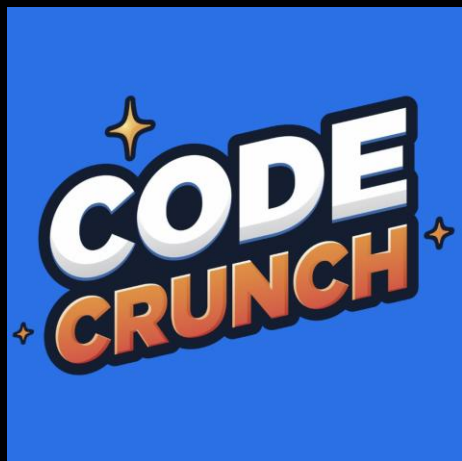
To **combine** the parts, we are going to use a **Revolute Mate**, allowing the blades to **connect and rotate** on the pole. Combine the **Mate Connector** of the hole of **Part 2** with the **Mate Connector** of the stick of **Part 1**.

The **windmill** is now completed!





Q&A Session



Attendance



Join us for the upcoming weekly sessions this Spring 2025!
(Jan 6 - Apr 4, 2025)



Monday: C5 1PM
Tuesdays: C7 1PM / C11 3PM
Wednesdays: C5 12:30 PM / C9 1:30 PM (Varies)
C1 2:30PM / C2 3:30PM / C13 5:30PM
Thursdays: C6 1PM / C8 3PM
Friday: C9 2PM (Varies) / C10 3PM/ C11 4PM



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